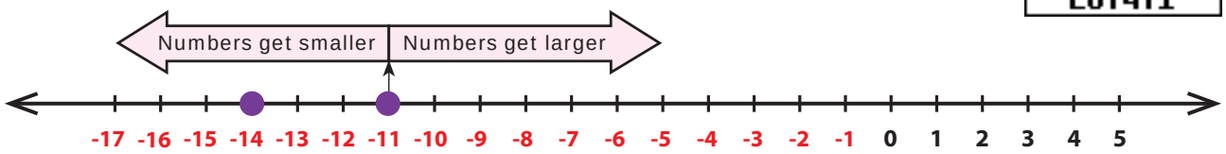
Example 6 Compare −14 and −11

E6T4Y1

Solution

Draw number line and plot the numbers −14 and −11 as follows.



Fixing −11, we find −14 is to the left of −11. So, −14 is smaller than −11. That is, −14 < −11.

Think

For two numbers, say 3 and 5, we know that 5 > 3. Will there be a change in the inequality if both the numbers have negative sign before them?

Activity

Take two cards from a deck of playing cards and identify, which is greater between them, assuming that the Joker card represents zero, black cards represent positive integers, red cards represent negative integers and the cards A, J, Q and K represent 1,11,12 and 13 respectively.

Example 7 Arrange the following integers in ascending order:

−15, 0, −7, 12, 3, −5, 1, −20, 25, 18

Solution

Step 1: First, separate the positive integers as 12, 3, 1, 25, 18 and the negative integers as −15, −7, −5, −20

Step 2: We can easily arrange positive integers in ascending order as 1, 3, 12, 18, 25 and negative integers in ascending order as −20, −15, −7, −5.

Step 3: As 0 is neither positive nor negative, it stays at the middle and now the arrangement −20, −15, −7, −5, 0, 1, 3, 12, 18 and 25 is in ascending order.

Try these

i) Is −15 < −26? Why?
ii) Which is smaller −3 or −5? Why?
iii) Which is greater 7 or −4? Why?
iv) Which is the greatest negative integer?
v) Which is the smallest positive integer?

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